Chapter 9. Bootstrap Tests

Parametric and nonparametric bootstrap tests, duality with confidence intervals

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Overview

The bootstrap approximates an unknown population by the given sample to approximate the sampling distribution of a statistic or estimator: for a population F and a parameter $\theta = \theta(F)$, it approximates the distribution $P(\hat{\theta}_n \le x \mid F)$ of the statistic $\hat{\theta}_n$ by bootstrap resampling, and this bootstrap sampling distribution generally depends on the actual value of θ. In contrast, in a hypothesis test comparing $H_0 : \theta = \theta_0$ with $H_1 : \theta \ne \theta_0$, the interest is the distribution of the statistic $\hat{\theta}_n$ when the null is true, i.e. when $\theta = \theta_0$ (the null distribution). In Part I we studied permutation tests, which obtain the null distribution of a test statistic by shuffling the observations in various situations.

Can the null distribution be approximated by bootstrap resampling? The bootstrap distribution approximates the actual population distribution F, so is it impossible when the population F_0 under the null differs from the actual F? Under a completely nonparametric model (no restriction on the form of the population), bootstrap approximation under F_0 is generally impossible. This chapter confirms that in some special situations, particularly semiparametric models with some modeling added, the null distribution of a statistic can be approximated by the bootstrap, and looks at test methods using this.

To aid understanding, we first look at bootstrap tests under a parametric model (e.g. the normal model), then explain bootstrap tests in nonparametric or semiparametric models with examples of tests about the location of one population and comparing two populations.[1]

9.1 Parametric bootstrap tests

A parametric bootstrap test is a test using parametric bootstrap resampling under the assumption that the population of the observations follows some parametric model (e.g. the normal family, the exponential family). Since a parametric model is assumed, one might think the bootstrap is unnecessary, but it is a test method used a lot in practice.

9.1.1 Resampling with a pivot

To explain the idea, assume observations $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ in the familiar normal model, and consider testing $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$, with the variance σ^2 unknown. The optimal test is the t-test with the statistic

$$T_0 = \frac{\bar{X}_n - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \quad (\text{when } H_0 : \mu = \mu_0 \text{ is true}). \tag{9.1}$$

Since T_0 has the form of a pivot, even when the null is not true, e.g. when $\mu = \mu_1$,

$$T_1 = \frac{\bar{X}_n - \mu_1}{s/\sqrt{n}} \sim t_{n-1} \quad (\text{when } \mu = \mu_1 \text{ is true}) \tag{9.2}$$

follows the same t_{n-1} distribution. As explained in Section 7.4, parametric bootstrap resampling substitutes the estimates $\hat{\mu} = \bar{X}_n$, $\hat{\sigma}^2 = s^2$ of the parameters μ, σ^2 and repeatedly generates bootstrap samples $\mathbb{X}_n^* = \{X_1^*, \dots, X_n^*\}$ from the bootstrap "population" $F^* = N(\hat{\mu}, \hat{\sigma}^2)$. Then

$$X_1^*, \dots, X_n^* \overset{\text{i.i.d.}}{\sim} N(\hat{\mu}, \hat{\sigma}^2),$$

so with the bootstrap-replicated sample mean $\bar{X}_n^* = \frac{1}{n}\sum_{i=1}^n X_i^*$ and sample standard deviation

$$s^* = \sqrt{\frac{1}{n-1}\sum_{i=1}^n (X_i^* - \bar{X}_n^*)^2},$$

[1] Bootstrap tests can also be used in more complex situations including regression; bootstrap tests in regression are treated in Chapter 10. For more general bootstrap test methods see Davison and Hinkley (1997).

the t-statistic is

$$T^* = \frac{\bar{X}_n^* - \hat{\mu}}{s^*/\sqrt{n}} \sim t_{n-1}. \tag{9.3}$$

Note that the distributions of (9.1), (9.2) and (9.3) all coincide. Since the test statistic T_0 is a pivot, its null distribution equals the sampling distribution of the pivot $T = (\bar{X}_n - \mu)/(s/\sqrt{n})$, where μ is the population mean μ of the actual population. This sampling distribution can be bootstrap-approximated.[2] The key of the parametric bootstrap test is to approximate this sampling distribution (the null distribution of T) by the bootstrap sampling distribution, i.e. the distribution of the replicates T^*, using the parametric bootstrap.

9.1.2 Resampling under the null hypothesis

When a pivot containing the test statistic for the given hypothesis is hard to find in a given parametric model, a statistic distinguishing null and alternative (even if not a pivot) can still be used. Continue assuming a normal population $N(\mu, \sigma^2)$ and testing $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$. Even without deep knowledge of the normal distribution, the statistic $D = \bar{X}_n - \mu_0$ obviously tends to be larger when the alternative is true, so take it as test statistic. The null distribution of this (roughly chosen) D is

$$D = \bar{X}_n - \mu_0 \sim N(0, \sigma^2) \quad (\text{when } H_0 : \mu = \mu_0 \text{ is true}). \tag{9.4}$$

(More precisely $N(0, \sigma^2/n)$.) Accepting the null $H_0 : \mu = \mu_0$, the distribution of the observations is $X_i \sim N(\mu_0, \sigma^2)$. If the still unknown σ^2 is estimated by its maximum likelihood estimate under the null,

$$\hat{\sigma}_0^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu_0)^2,$$

the most plausible "population" under the null is $F_0^* = N(\mu_0, \hat{\sigma}_0^2)$, and parametric bootstrap resampling can be done from this distribution F_0^*. With bootstrap samples $\{X_1^{0*}, \ldots, X_n^{0*}\} \overset{\text{i.i.d.}}{\sim} N(\mu_0, \hat{\sigma}_0^2)$, the bootstrap replicates are

$$D^* = \bar{X}_n^{0*} - \mu_0 \sim N(0, \hat{\sigma}_0^2/n). \tag{9.5}$$

When $\hat{\sigma}_0^2 \approx \sigma^2$ (when the null is true and the sample is large), the distribution (9.5) of D^* is expected to approximate the null distribution (9.4) of D well. So taking the distribution of D^* generated by the bootstrap as the null distribution of D is the bootstrap test with resampling under the null.

If the pivot T_0 (9.1) is taken as the test statistic, the bootstrap replicates under resampling under the null (here the bootstrap sample is $X_i^{0*} \sim N(\mu_0, \hat{\sigma}_0^2)$) are

$$T_0^* = \frac{\bar{X}_n^{0*} - \mu_0}{s^{0*}/\sqrt{n}} \sim t_{n-1}, \tag{9.6}$$

so the (bootstrapped) distribution of T_0^* equals the distribution t_{n-1} (9.3) of the pivot-resampled T^*. (Supplied, Exercise 9.1 of the book: under $X_i^{0*} \overset{\text{i.i.d.}}{\sim} N(\mu_0, \hat{\sigma}_0^2)$, T_0^* is the one-sample t-statistic of a normal sample with true mean μ_0, so it is exactly t_{n-1} whatever $\hat{\sigma}_0^2$.) I.e. when the test statistic is a pivot, resampling under the null and resampling with the pivot are approximately equivalent. When the test statistic is a pivot the bootstrap approximation is more accurate, so it is better to take a (or an approximately) pivotal test statistic. But when a pivot for the parameter to be tested is hard to find, resampling under the null may be the only bootstrap test method.

[2] Under normality all distributions are t-distributions, so the bootstrap is not needed in this example. The bootstrap approximation is used when the sampling distribution of the statistic T is unknown.

9.1.3 Bootstrap tests in general parametric models

Summarizing the two methods explained for the normal distribution for a general parametric model: in a parametric model $F(\cdot;\theta,\eta)$ we test the null $H_0:\theta=\theta_0$. The parameter θ is the one whose value the null fixes, and η the parameter the null does not fix. E.g. in the normal model $N(\mu,\sigma^2)$ with $\theta=\mu$, $\eta=\sigma^2$. If an exponential model $\text{Exp}(\theta)$ is assumed and a hypothesis about the mean set, there is no parameter η, so it can be ignored. Given data $\mathbb{X}_n=\{x_1,\ldots,x_n\}$, the test statistic is taken as a pivot (or approximate pivot) $S=S(\mathbb{X}_n,\theta_0)$.[3] (Supplied check: for Laplace(μ,b), $|X_i-\mu|/b\sim\text{Exp}(1)$, so $2\sum_i|X_i-\mu|/b\sim\chi^2_{2n}$; with $b=2$ this is $\sum_i|X_i-\mu|$.) The test rejects the null when the observed value S_{obs} of S is large.

1. **Resampling under the null:** assume the null $\theta=\theta_0$ is true, find from the data the maximum likelihood estimate $\hat\eta_0$ of the unknown parameter η, carry out parametric bootstrap resampling from $F(\cdot;\theta_0,\hat\eta_0)$, and obtain the bootstrap replicates $S_0^{*(b)}=S(\mathbb{X}_n^{*(b)},\theta_0)$ of the statistic S. The null distribution of S is taken as the distribution of $\{S_0^{*(b)}:b=1,\ldots,B\}$, and the p-value computed:

$$p_0=\frac{1+\sum_{b=1}^B I(S_0^{*(b)}\geq S_{\text{obs}})}{B+1}.$$

2. **Resampling with a pivot:** obtain from the given data the maximum likelihood estimates $\hat\theta,\hat\eta$, carry out parametric bootstrap resampling from $F(\cdot;\hat\theta,\hat\eta)$, and obtain the bootstrap replicates $S^{*(b)}=S(\mathbb{X}_n^{*(b)},\hat\theta)$ of the statistic S. ($\hat\theta$ is used in bootstrap resampling instead of θ_0.) Since S is a pivot, its null distribution is approximated by the distribution of $\{S^{*(b)}:b=1,\ldots,B\}$. So the p-value is

$$p_{\text{pivot}}=\frac{1+\sum_{b=1}^B I(S^{*(b)}\geq S_{\text{obs}})}{B+1}.$$

Two examples compare the resampling methods and the choice of test statistic.

Parametric bootstrap test for the location of one population. Using the gene expression levels $x_1,\ldots,x_n$ of the $n=25$ AML patients among the leukemia data seen in Example 4.1 of Chapter 4, we test the hypothesis $H_0:\theta=166$ against $H_1:\theta>166$ about the location (mean) θ of this population. Figure 9.1 of the book shows these data.

Assume the variation of gene expression in this population can be modeled by the Laplace distribution, i.e. $X_i\overset{\text{i.i.d.}}{\sim}\text{Laplace}(\theta,b)$. Then the maximum likelihood estimates from the data are $\hat\theta=\text{median}\{X_i\}$, $\hat b=\frac{1}{n}\sum_{i=1}^n|X_i-\hat\theta|$, and under the null ($\theta=\theta_0=166$) the maximum likelihood estimate of b is $\hat b_0=\frac{1}{n}\sum_{i=1}^n|X_i-\theta_0|$. For the following three test statistics we carry out bootstrap tests with the two kinds of resampling (six bootstrap tests in all):

- t-statistic: $T=(\bar X_n-\theta_0)/(s/\sqrt{n})$;
- difference of means: $D=\bar X_n-\theta_0$;
- ratio of medians: $M=\text{median}\{X_i\}/\theta_0=\hat\theta/\theta_0$.

The observed values of the statistics:

[3] The pivot for μ in the normal $N(\mu,\sigma^2)$ is $S(\mathbb{X}_n,\mu_0)=(\bar X_n-\mu_0)/(s/\sqrt{n})\sim t_{n-1}$, and the pivot for σ^2 is $S(\mathbb{X}_n,\sigma_0^2)=(n-1)s^2/\sigma_0^2\sim\chi^2_{n-1}$. Assuming a parametric model does not mean a pivot can always easily be found. In the Laplace distribution Laplace$(\mu,2)$ the pivot for μ is $S(\mathbb{X}_n,\mu_0)=\sum_{i=1}^n|X_i-\mu_0|\sim\chi^2_{2n}$, but in Laplace$(\mu,b)$ a pivot for μ is not easy to find. Then the approximate pivot $S(\mathbb{X}_n,\mu_0)=(\bar X_n-\mu_0)/(s/\sqrt{n})\overset{\cdot}{\sim}N(0,1)$ from the CLT can be used.

```
> leukemia <- read.csv("data/leukemia4.csv")
> x <- leukemia$ge[leukemia$label == "AML"]
> # hypothesis H0: mu =mu0, H1: mu > mu0
> mu0 = 166
> # test statistic (two test stats)
> t0 = sqrt(n) * (mean(x) - mu0 ) / sd(x)
> d0 = mean(x) - mu0
> m0 = median(x) / mu0
> c(t0,d0,m0)
[1]  2.722207 67.120000  1.451807
```

Resampling under the null. Bootstrap samples are drawn from Laplace$(\theta_0, \hat{b}_0)$, and from each the mean $\bar{X}_n^{0*}$, standard deviation s^{0*} and median median$\{X_i^{0*}\}$ are computed; the distributions of the bootstrap replicates

$$T_0^* = (\bar{X}_n^{0*} - \theta_0)/(s^{0*}/\sqrt{n}), \qquad D_0^* = \bar{X}_n^{0*} - \theta_0, \qquad M_0^* = \text{median}\{X_i^{0*}\}/\theta_0 \tag{9.7}$$

are taken as the null distributions of the respective statistics. (In the superscript "$0*$" of statistics computed from bootstrap samples, "0" means a sample under the null model and "$*$" a bootstrap sample.)

Resampling with a pivot. The method proceeds assuming each statistic is an approximate pivot. (By the CLT, T and D are easily seen to be approximate pivots, but $M(=\text{median}\{X_i\}/\theta)$ has asymptotically mean 1 and variance proportional to θ^{-2}, so it is not even an approximate pivot.) Bootstrap samples are drawn from Laplace$(\hat{\theta}, \hat{b})$, and from each the mean $\bar{X}_n^*$, standard deviation s^* and median median$\{X_i^*\}$ are computed; the distributions of the bootstrap replicates

$$T^* = (\bar{X}_n^* - \hat{\theta})/(s^*/\sqrt{n}), \qquad D^* = \bar{X}_n^* - \hat{\theta}, \qquad M^* = \text{median}\{X_i^*\}/\hat{\theta} \tag{9.8}$$

are taken as the null distributions. The bootstrap-approximated null distributions in each case are obtained by:

```
# assume Laplace dist'n, and get MLEs
mu.hat = median(x); b.hat = mean(abs(x - mu.hat)) # mle (general)
b.hat0 = mean(abs(x-mu0))  # mle under H0

# parametric bootstrap for (T, D, M)
set.seed(1)
boot.out = replicate(3999, {
  x.boot = rlaplace(n, mu = mu.hat, sigma = b.hat) # sample under the
    best-fitting model
  x.boot0 = rlaplace(n, mu = mu0, sigma = b.hat0) # sample under null
    model
  c(mean(x.boot), sd(x.boot), mean(x.boot0), sd(x.boot0), median(x.
    boot), median(x.boot0))
}) %>% t()

t.boot0 = sqrt(n) * (boot.out[,3] - mu0) / boot.out[,4]
t.boot = sqrt(n) *(boot.out[,1] - mu.hat) / boot.out[,2]
d.boot0 = boot.out[,3] - mu0
d.boot = boot.out[,1] - mu.hat
m.boot0 = boot.out[,6] / mu0
m.boot = boot.out[,5] / mu.hat
```

Each null distribution is compared with the observed statistic to get the p-value. Figure 9.2 of the book shows the bootstrap-approximated null distributions and p-values in all six cases; every test chose the alternative $\theta > 166$ at the 5% level (p-values: T 0.0035 / 0.0055, D 0.0205 / 0.0145, M 0.00475 / 0.00025 for null / pivot resampling). Comparing the test methods:

1. The two resampling methods do not give very different results. But the statistic M is not a pivot, so "pivot" resampling is not recommended for it.
2. The p-value of the bootstrap test with D is higher than with the other statistics. As under normality, the test using the studentized T is more powerful than the test using D. Building test statistics in the "studentized difference" form like T was recommended by Hall and Wilson (1991).
3. The median, the maximum likelihood estimator of the parameter θ, is the best estimator of the parameter, so the test using the median, M, is more powerful. For reference, taking the more familiar form $M' = \text{median}\{X_i\} - \theta_0$ as statistic, the result with resampling under the null is very similar to that with M, but the result with pivot resampling differs slightly. (Here M' is a pivot.)
4. Since the (possibly wrong) assumption that the population follows the Laplace distribution was made, there is no guarantee that every test controls the type I error. Assuming a more appropriate parametric model than the Laplace (e.g. the normal model) controls the type I error more exactly while raising the power.

Hypothesis test for the locations of two populations. Using all the gene expression data of the leukemia patients, we run a parametric bootstrap test for the location difference of the two populations. Take the expression levels of the $m = 47$ ALL patients as observations $X_1, \ldots, X_m$ of the first population and those of the $n = 25$ AML patients as observations $Y_1, \ldots, Y_n$ of the second, and assume the parametric model

$$X_i \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\mu_1, b), \qquad Y_i \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\mu_2, b). \tag{9.9}$$

We bootstrap-test the null $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 < \mu_2$, or $H_0 : \theta = 0$ against $H_1 : \theta > 0$ with $\theta = \mu_2 - \mu_1$.

Write the maximum likelihood estimates of the parameters of model (9.9) as $\hat{\mu}_1 = \text{median}\{X_i\}$, $\hat{\mu}_2 = \text{median}\{Y_i\}$, $\hat{b} = \frac{1}{m+n}\big(\sum_{i=1}^m |X_i - \hat{\mu}_1| + \sum_{i=1}^n |Y_i - \hat{\mu}_2|\big)$, and under the null $\hat{\mu}_0 := \hat{\mu}_1^{(0)} = \hat{\mu}_2^{(0)} = \text{median}\{X_i, Y_j\}$ (the median of all observations), $\hat{b}_0 = \frac{1}{m+n}\big(\sum_{i=1}^m |X_i - \hat{\mu}_0| + \sum_{i=1}^n |Y_i - \hat{\mu}_0|\big)$. Both parameters μ_1, μ_2 are medians and means, so we take the following two statistics as test statistics and consider bootstrap tests with the two kinds of resampling:

- studentized difference of means: $T = (\bar{Y}_n - \bar{X}_m)/\widehat{\text{se}} = (\bar{Y}_n - \bar{X}_m - \theta_0)/\widehat{\text{se}}$ (here $\widehat{\text{se}}$ is the standard error estimate of $\bar{Y}_n - \bar{X}_m$ assuming equal variances, and θ_0 the value of θ under the null, $\theta_0 = 0$);
- difference of medians: $M = \text{median}\{Y_i\} - \text{median}\{X_i\} = \text{median}\{Y_i\} - \text{median}\{X_i\} - \theta_0$.

Resampling under the null. When the null is true, all observations were sampled at random from $\text{Laplace}(\mu_0, b)$, so bootstrap samples are drawn as

$$X_1^{0*}, \ldots, X_m^{0*}, Y_1^{0*}, \ldots, Y_n^{0*} \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\hat{\mu}_0, \hat{b}_0).$$

(Here under the null the two populations are not distinguished; if the scale b of the Laplace distribution differed by population, the bootstrap samples would be drawn from each sample.) From these bootstrap samples the replicates

$$T_0^* = (\bar{Y}_n^{0*} - \bar{X}_m^{0*})/s_p^{0*}, \qquad M_0^* = \text{median}\{Y_i^{0*}\} - \text{median}\{X_i^{0*}\}$$

are computed, and their distributions used as the null distributions of the statistics to compute p-values.

Resampling with a pivot. Both statistics are approximate pivots, so pivot resampling proceeds as follows. With the estimates $(\hat{\mu}_1, \hat{\mu}_2, \hat{b})$ of each population's parameters, bootstrap samples

$$X_1^*, \ldots, X_m^* \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\hat{\mu}_1, \hat{b}), \qquad Y_1^*, \ldots, Y_n^* \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\hat{\mu}_2, \hat{b})$$

are drawn, and the distributions of the replicates

$$T_0^* = (\bar{Y}_n^* - \bar{X}_m^* - (\hat{\mu}_2 - \hat{\mu}_1))/s_p^*, \qquad M_0^* = \text{median}\{Y_i^*\} - \text{median}\{X_i^*\} - (\hat{\mu}_2 - \hat{\mu}_1)$$

are used as the null distributions of the statistics to compute p-values. The R code implementing this follows; the function `pooled.se()` is defined in `np_supp.R`.

```
source("np_supp.R")

leukemia <- read.csv("data/leukemia4.csv")
y = leukemia$ge[leukemia$label == "AML"]; m = length(x)
x = leukemia$ge[leukemia$label == "ALL"]; n = length(y)

# test statistic (two test stats)
t0 = (mean(y) - mean(x)) / pooled.se(x,y) #t.test(y,x, var.equal = T)$
    statistic
m0 = median(y) - median(x)

# assume Laplace, get MLE
mu1.hat = median(x); mu2.hat = median(y)
b.hat = mean( abs( c(x - mu1.hat, y - mu2.hat))) # mle (general)
mu0.hat = median(c(x,y))
b.hat0 = mean(abs(c(x,y) - mu0.hat))  # mle under H0

# parametric bootstrap for both (T,M)
set.seed(1)
boot.out = replicate(3999, {
  x.boot = rlaplace(m, mu = mu1.hat, sigma = b.hat)
  y.boot = rlaplace(n, mu = mu2.hat, sigma = b.hat)

  x.boot0 = rlaplace(m, mu = mu0.hat, sigma = b.hat0)
  y.boot0 = rlaplace(n, mu = mu0.hat, sigma = b.hat0)

  t.boot0 = (mean(y.boot0) - mean(x.boot0)) / pooled.se(x.boot0,y.
    boot0)
  t.boot = (mean(y.boot) - mean(x.boot) - (mu2.hat - mu1.hat)) /
    pooled.se(x.boot,y.boot)

  m.boot0 = median(y.boot0) - median(x.boot0)
  m.boot = median(y.boot) - median(x.boot)- (mu2.hat - mu1.hat)

  c(t.boot0, t.boot, m.boot0, m.boot)
}) %>% t()
```

(In the book's code, `m` and `n` are assigned before `x` is redefined; with $m = \#\text{ALL} = 47$, $n = \#\text{AML} = 25$ the lengths should be `m = length(x)` after defining `x` as ALL and `n = length(y)`.) The null distributions and p-values in each case are in Figure 9.3 of the book (T: 0.0095 / 0.007; M: 0.0162 / 0.016 for null / pivot resampling). All tests chose the alternative $H_1 : \mu_1 < \mu_2$ significantly at the 5% level. (The two-sample t-test, the permutation test with the t-statistic, and the distribution-free Wilcoxon–Mann–Whitney test give the same result.)

As in the one-population case, the two resampling methods hardly differ. Resampling with a pivot is recommended by Hall and Wilson (1991) among others, but in bootstrap tests in complex parametric models, resampling under the null may be the only feasible bootstrap method.

9.2 Nonparametric bootstrap tests

A nonparametric or distribution-free test is a test whose performance (control of the type I error and high power) is guaranteed under the assumption of a broad family of populations. The permutation test, which obtains the null distribution of a carefully defined statistic by shuffles, is the representative nonparametric test. Permutation tests have the disadvantage that they must satisfy the exchangeability assumption and cannot be applied when it fails. If the permutation test is understood as a resampling method that samples the observations without replacement, one can consider applying bootstrap sampling (sampling with replacement) instead of sampling without replacement.

A simple example first. With observations from two populations $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F_1$, $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} F_2$, we test the hypotheses

$$H_0 : F_1 = F_2 \quad \text{vs} \quad H_1 : F_1 \neq F_2. \tag{9.10}$$

If the locations of the two populations are expected to differ, the two-sample t-statistic or the Mann–Whitney U-statistic can be the test statistic; more generally the Kolmogorov–Smirnov statistic K can be used, and tests by permutation or large-sample approximation were treated in Chapter 3. Can bootstrap resampling be used instead of resampling by shuffles? The method of resampling under the null seen above can be applied to this situation.

Under the null, all observations were sampled from one population $F = F_1 = F_2$, so bootstrap resampling assuming this null is done by random sampling from the empirical distribution approximating this common population F,

$$F_N(x) = \frac{1}{m+n}\Big\{\sum_{i=1}^{m} I(X_i \le x) + \sum_{i=1}^{n} I(Y_i \le x)\Big\}, \tag{9.11}$$

or by sampling $N = m + n$ data with replacement from $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$. The distribution of the test statistic bootstrap-resampled in this way is the null distribution of that statistic. (Unless the samples are too small, the results of the permutation test and of this bootstrap test are similar.)

Example 9.1

With the leukemia data, carry out the nonparametric bootstrap test of hypotheses (9.10) using (9.11). The bootstrap samples $x_1^*, \ldots, x_m^*$ are drawn at random from F_N (9.11), and $y_1^*, \ldots, y_n^*$ also from F_N; so $x_1^*, \ldots, x_m^*, y_1^*, \ldots, y_n^* \overset{\text{i.i.d.}}{\sim} F_N$. The R code shows the bootstrap test with the two-sample t-statistic:

```
> t0 = (mean(y) - mean(x)) / pooled.se(x,y)
> set.seed(1)
> B = 3999
> boot.out = replicate(B, {
+   boot.id = sample(1:(m+n),(m+n),replace = TRUE)
+   x.boot = c(x,y)[boot.id[1:m]]
+   y.boot = c(x,y)[boot.id[(m+1):(m+n)]]
+   (mean(y.boot) - mean(x.boot)) / pooled.se(x.boot,y.boot)
+ })
> boot.p = 2* min( c( mean(c(t0,boot.out) >=t0),
+                     mean(c(t0,boot.out) <=t0)))
> boot.p
[1] 0.011
```

The bootstrap p-value is 0.011. For comparison, the permutation test (with the two-sample t-statistic) on the same data gives a random permutation p-value of 0.016: the bootstrap test and the permutation test reach the same conclusion. In this situation the only difference between the permutation test and the bootstrap test procedures is: resampling the observa-

tions without replacement gives the permutation test, with replacement the bootstrap test. Compare the procedures and results in the R code below.

```
> perm.out = replicate(B, {
+   perm.id = sample(1:(m+n),(m+n),replace = FALSE)
+   x.perm = c(x,y)[perm.id[1:m]]
+   y.perm = c(x,y)[perm.id[(m+1):(m+n)]]
+   (mean(y.perm) - mean(x.perm)) / pooled.se(x.perm,y.perm)
+ })
> perm.p = 2* min( c( mean(c(t0,perm.out) >=t0),
+                     mean(c(t0,perm.out) <=t0)))
> perm.p
[1] 0.016
```

If we want to test the location difference of two populations, the null of (9.10) (the two distributions are completely identical) may be too narrow. E.g. to allow the dispersions of the two populations to differ even when the null of equal locations is true, neither the ordinary permutation test nor the bootstrap test just explained can be used. Bootstrap resampling is possible within each population, unlike the permutation test, which shuffles the two populations, so tests by bootstrap resampling are possible even when the dispersions of the two populations differ. For this a semiparametric model, applicable even under the null, is needed.

We look at nonparametric bootstrap tests using a representative semiparametric model for hypothesis tests about the locations of populations.

9.2.1 Bootstrap tests for the locations of two populations

For the gene expression data of leukemia patients, when a parametric model is hard to assume, the following semiparametric location model can be assumed:

$$X_i = \mu_1 + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F, \quad i = 1, \ldots, m,$$
$$Y_i = \mu_2 + e_i, \qquad e_i \overset{\text{i.i.d.}}{\sim} G, \quad i = 1, \ldots, n.$$

Here F and G are distributions with mean or median 0 and need not be equal. The location shift $\Delta = \mu_2 - \mu_1$ of the two populations is defined regardless of the unknown distributions F, G. We are interested in testing the hypotheses about the locations of the two populations

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_2 > \mu_1. \tag{9.12}$$

(The book prints $H_1 : \mu_1 > \mu_2$; the test below and Example 9.2 are upper-tailed in $\Delta = \mu_2 - \mu_1$, i.e. $H_1 : \Delta > 0$.) (Even when the null is true the two populations may differ, so the null of (9.12) contains more distributions than the null of (9.10).)

If the distributions F, G have known forms, the parametric bootstrap test explained above can be used. E.g. if $X_i \sim \text{Laplace}(\mu_1, b_1)$ and $Y_i \sim \text{Laplace}(\mu_2, b_2)$, the (approximate) pivot for $\Delta = \mu_2 - \mu_1$,

$$T = T_\Delta = \frac{\bar{Y}_n - \bar{X}_m - \Delta}{\sqrt{\frac{1}{m}s_1^2 + \frac{1}{n}s_2^2}}, \tag{9.13}$$

can be used. Substitute the null value $\Delta = \Delta_0 = 0$ for the parameter Δ in (9.13) and the observations for X_i, Y_i to get the observed T_{obs} as test statistic, and obtain the null distribution of this statistic by pivot resampling.

Even in the nonparametric situation where the forms of F, G cannot be specified, assuming the statistic T of (9.13) is an approximate pivot, the nonparametric bootstrap can be used. Writing the estimate of Δ as $\hat{\Delta} = \bar{Y}_n - \bar{X}_m$ and its standard error $\widehat{\text{se}} = \sqrt{\frac{1}{m}s_1^2 + \frac{1}{n}s_2^2}$, $T = (\hat{\Delta} - \Delta)/\widehat{\text{se}}$.

Nonparametric resampling with a pivot. The approximate distribution H of the pivot T is bootstrap-approximated as

$$\begin{aligned} T &= \frac{\bar{Y}_n - \bar{X}_m - \Delta}{\sqrt{\frac{1}{m}s_1^2 + \frac{1}{n}s_2^2}} = \frac{\hat{\Delta} - \Delta}{\widehat{\mathrm{se}}} \dot{\sim} H \\ \overset{d}{\approx} T^* &= \frac{\bar{Y}_n^* - \bar{X}_m^* - (\bar{Y}_n - \bar{X}_m)}{\sqrt{\frac{1}{m}(s_1^*)^2 + \frac{1}{n}(s_2^*)^2}} = \frac{\hat{\Delta}^* - \hat{\Delta}}{\widehat{\mathrm{se}}^*} \dot{\sim} H^*. \end{aligned} \tag{9.14}$$

Here the bootstrap samples are: $x_1^*, \ldots, x_m^*$ drawn as m simple random samples with replacement from the observations $\{X_1, \ldots, X_m\}$, and $y_1^*, \ldots, y_n^*$ as n from $\{Y_1, \ldots, Y_n\}$. Since the distributions of the two populations differ under both null and alternative, bootstrap resampling is done separately. The distribution of the null test statistic T_0 is approximated by H, which is again bootstrap-approximated by H^*; so the nonparametric bootstrap test proceeds as follows (Efron and Tibshirani, 1994):

1. Draw $x_1^*, \ldots, x_m^*$ as m simple random samples with replacement from $\{X_1, \ldots, X_m\}$ and $y_1^*, \ldots, y_n^*$ as n from $\{Y_1, \ldots, Y_n\}$, and compute the bootstrap replicate T^*.
2. Repeat step 1 a predetermined number B of times, obtaining T^{*b} for $b = 1, \ldots, B$.
3. The p-value for hypotheses (9.12) is $\hat{p} = (1 + \sum_{b=1}^{B} I\{T^{*b} \geq T_{\mathrm{obs}}\})/(B+1)$.

That this p-value is approximately exact can be seen from:

$$p = P(T_{\Delta_0} \geq T_{\mathrm{obs}} \mid H_0) = P(T_\Delta \geq T_{\mathrm{obs}}) \approx P(T^* \geq T_{\mathrm{obs}}) \approx \hat{p}.$$

The first equality is the definition of the p-value and the second is given by the assumption that $T = T_\Delta$ is a pivot; the last two approximations are given by the bootstrap approximation.

Example 9.2

For hypotheses (9.12) on the leukemia data, carry out the nonparametric bootstrap test with the procedure of Efron and Tibshirani (1994). Comparing the null $H_0 : \Delta = \mu_2 - \mu_1 = 0$ with the alternative $H_1 : \Delta > 0$, obtain the null distribution of the statistic (9.13) by the nonparametric bootstrap and compute the p-value:

```
> d0 = mean(y) - mean(x)
> t0 = (mean(y) - mean(x)) / pooled.se(x,y)
> set.seed(1)
> B = 3999
> boot.out = replicate(B, {
+   x.boot = sample(x, m, replace = TRUE)
+   y.boot = sample(y, n, replace = TRUE)
+   t.boot = (mean(y.boot)  - mean(x.boot) - d0) / pooled.se(x,y
    )
+ })
> boot.p = mean(c(t0,boot.out) >= t0)
> boot.p
[1] 0.0065
```

(The book's code standardizes T^* with `pooled.se(x,y)` of the original data rather than the bootstrap sample; (9.14) uses $\widehat{\mathrm{se}}^*$.) Compared with the p-value of Example 9.1, the p-value using the pivot is slightly smaller. When (9.14) holds approximately, the bootstrap test using the pivot is more powerful.

Remark

When the distributions of the two populations differ, resampling under the null is not recommended. Suppose the null $\Delta = \mu_2 - \mu_1 = 0$ is true; using the estimate $\hat{\mu}_0 = (m\bar{X}_m + n\bar{Y}_n)/(m+n)$ of $\mu_0 = \mu_1 = \mu_2$, consider resampling from each sample $X_i - \hat{\mu}_0$ and $Y_i - \hat{\mu}_0$. Then the bootstrap replicates of T are

$$T_0^* = \frac{\bar{Y}_n^* - \bar{X}_m^* - (\hat{\mu}_0 - \hat{\mu}_0)}{\sqrt{\frac{1}{m}(s_1^*)^2 + \frac{1}{n}(s_2^*)^2}} = \frac{\bar{Y}_n^* - \bar{X}_m^*}{\sqrt{\frac{1}{m}(s_1^*)^2 + \frac{1}{n}(s_2^*)^2}},$$

and the distribution of the T_0^* is the bootstrap sampling distribution of $T_0 = (\bar{Y}_n - \bar{X}_m)/\sqrt{\frac{1}{m}s_1^2 + \frac{1}{n}s_2^2}$, not its null distribution. (Centering at a common $\hat{\mu}_0$ does not remove the observed difference $\bar{Y}_n - \bar{X}_m$ from the resampled data, so T_0^* is centered near T_0 rather than near 0.)

9.2.2 Bootstrap tests for the location of one population

To test the location parameter of one population, assume the observations $X_1, \ldots, X_n$ follow the semiparametric location model

$$X_i = \theta + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F, \quad i = 1, \ldots, n.$$

With $\hat{\theta}_n$ an estimator of the location parameter θ, if $\hat{\theta}_n - \theta$ or the studentized $(\hat{\theta}_n - \theta)/\widehat{\text{se}}$ is a pivot, the distribution of this pivot can be bootstrap-approximated as

$$H^* \sim \hat{\theta}_n^* - \hat{\theta}_n \overset{d}{\approx} \hat{\theta}_n - \theta \sim H \tag{9.15}$$

or

$$H^* \sim \frac{\hat{\theta}_n^* - \hat{\theta}_n}{\widehat{\text{se}}^*} \overset{d}{\approx} \frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} \sim H. \tag{9.16}$$

Here $\hat{\theta}_n^*$ and $\widehat{\text{se}}^*$ are bootstrap replicates obtained by nonparametric bootstrap resampling, as for the bootstrap-t interval.

Suppose we test the null $H_0 : \theta = \theta_0$ against the alternative $H_1 : \theta > \theta_0$. With (9.15) the rejection region is $\hat{\theta}_n > c$ for some value c, so the p-value is

$$\begin{aligned} p_1 &= P(\hat{\theta}_n \geq \hat{\theta}_{\text{obs}} \mid H_0) \\ &= P(\hat{\theta}_n - \theta_0 \geq \hat{\theta}_{\text{obs}} - \theta_0 \mid H_0) \\ &= P(\hat{\theta}_n - \theta \geq \hat{\theta}_{\text{obs}} - \theta_0) \qquad \text{(for every } \theta, \text{ since } \hat{\theta}_n - \theta \text{ is a pivot)} \\ &\approx P(\hat{\theta}_n^* - \hat{\theta}_{\text{obs}} \geq \hat{\theta}_{\text{obs}} - \theta_0) \\ &\approx \frac{1 + \sum_{b=1}^{B} I(\hat{\theta}_n^{*b} - \hat{\theta}_{\text{obs}} \geq \hat{\theta}_{\text{obs}} - \theta_0)}{B+1} =: \hat{p}_1. \end{aligned} \tag{9.17}$$

With (9.16) the rejection region is

$$\frac{\hat{\theta}_n - \theta_0}{\widehat{\text{se}}} > c$$

for some value c, so the p-value is

$$\hat{p}_2 = \frac{1 + \sum_{b=1}^{B} I\{(\hat{\theta}_n^{*b} - \hat{\theta}_{\text{obs}})/\widehat{\text{se}}^* \geq (\hat{\theta}_{\text{obs}} - \theta_0)/\widehat{\text{se}}\}}{B+1}.$$

Justification of $\hat{p}_2$ (supplied; Exercise 9.4 of the book). Repeat (9.17) with the studentized pivot. With $\widehat{\text{se}}$ computed from the observed data,

$$p_2 = P\Big(\frac{\hat{\theta}_n - \theta_0}{\widehat{\text{se}}} \geq \frac{\hat{\theta}_{\text{obs}} - \theta_0}{\widehat{\text{se}}_{\text{obs}}} \;\Big|\; H_0\Big) = P\Big(\frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} \geq \frac{\hat{\theta}_{\text{obs}} - \theta_0}{\widehat{\text{se}}_{\text{obs}}}\Big)$$

for every θ, since $(\hat{\theta}_n - \theta)/\widehat{\text{se}}$ is a pivot. By (9.16) this is approximately $P((\hat{\theta}^*_n - \hat{\theta}_{\text{obs}})/\widehat{\text{se}}^* \geq (\hat{\theta}_{\text{obs}} - \theta_0)/\widehat{\text{se}}_{\text{obs}})$, which the B bootstrap replicates estimate by $\hat{p}_2$. ■

If the alternative is two-sided, $H_1 : \theta \neq \theta_0$ (using (9.16)), the rejection region is $|\frac{\hat{\theta}_n - \theta_0}{\widehat{\text{se}}}| > c$, and the p-value (as in (9.17)) is

$$\hat{p}_{\text{both}} = \frac{1 + \sum_{b=1}^{B} I\{|(\hat{\theta}^{*b}_n - \hat{\theta}_{\text{obs}})/\widehat{\text{se}}^*| \geq |(\hat{\theta}_{\text{obs}} - \theta_0)/\widehat{\text{se}}|\}}{B+1}.$$

Example 9.3

As in Section 9.1.3, test $H_0 : \theta = 166$ against $H_1 : \theta > 166$ for the mean θ of the gene expression levels of the AML patients among the leukemia data. The p-value of the bootstrap test using the pivot (9.16) is computed as follows:

```
> x <- leukemia$ge[leukemia$label == "AML"]
> n = length(x)
>
> # hypothesis H0: mu =mu0, H1: mu > mu0
> mu0 = 166
>
> # test statistic
> hat.mu = mean(x)
> t0 = sqrt(n) * (mean(x) - mu0 ) / sd(x)
>
> # nonparametric bootstrap
> set.seed(1)
> boot.out = replicate(3999, {
+   x.boot = sample(x, n, replace = TRUE)
+   sqrt(n) * (mean(x.boot) - hat.mu) / sd(x.boot)
+ })
> boot.p = mean(c(t0,boot.out) >= t0)
> boot.p
[1] 0.01225
```

9.3 Duality of bootstrap tests and bootstrap confidence intervals

Bootstrap hypothesis tests and bootstrap confidence intervals are closely related. We know that in some parametric models the complement of the rejection region of a level-α test coincides with the level-$(1-\alpha)$ confidence interval: if a value θ_0 of a parameter lies in the 95% interval, the test of the null $H_0 : \theta = \theta_0$ about that parameter does not reject the null at level 5%. Conversely, if θ_0 does not lie in the interval, the test rejects the null.

Likewise, if the p-value of the two-sided test using the pivot (9.15) is larger than $\alpha = 0.05$ (fails to reject $H_0 : \theta = \theta_0$), the null value θ_0 lies in the 95% basic bootstrap interval constructed assuming (9.15) is a pivot. Conversely, if θ_0 is contained in (or not contained in) the 95% basic bootstrap interval for θ, the null $H_0 : \theta = \theta_0$ fails to be rejected (or is rejected) at level 5%. Similarly, the test using the pivot (9.16) corresponds to the bootstrap-t interval.

Sketch (supplied). Using (9.15), the two-sided level-α test fails to reject $H_0 : \theta = \theta_0$ iff $Q_{H^*}(\frac{\alpha}{2}) \leq \hat{\theta}_{\text{obs}} - \theta_0 \leq Q_{H^*}(1-\frac{\alpha}{2})$, i.e. iff $\hat{\theta}_{\text{obs}} - Q_{H^*}(1-\frac{\alpha}{2}) \leq \theta_0 \leq \hat{\theta}_{\text{obs}} - Q_{H^*}(\frac{\alpha}{2})$, which is the basic bootstrap interval (8.4). The same argument with the studentized pivot gives the bootstrap-t interval (8.10). ■

In the same context, a hypothesis test can be made by "inverting" the percentile interval (or adjusted percentile interval). The percentile interval was developed assuming the existence of some monotone transform, and since that monotone transform is unknown, a test method corresponding to the percentile interval was not developed separately.